

Arachnoid Membrane Segmentation in Intraoperative Microscopic MVD Surgery Scenes

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Abstract. Microvascular decompression (MVD) is a neurosurgical procedure to treat cranial nerve compression syndromes such as trigeminal neuralgia and hemifacial spasm. The arachnoid membrane (AM) is a thin, transparent meningeal layer that adheres to or covers neurovascular structures and must be carefully dissected to access the surgical site during MVD surgery. Proper AM dissection is essential for visualizing the operative field and ensuring safe vessel and nerve manipulation. However, AM dissection is technically challenging due to its poor contrast with surrounding tissues and close adherence to critical neurovascular structures. To address this, we propose the first dedicated study on AM segmentation from operative MVD videos. We introduce a high-quality, expert-annotated dataset focusing on AM in the cerebellopontine angle and train a segmentation model with a task-specific loss function to improve AM segmentation. Our results demonstrate that the proposed loss function improves AM segmentation performance by 7.35% in IoU over the baseline, enabling reliable segmentation despite the membrane’s transparency and intraoperative variability. This work lays the foundation for automated AM recognition in surgical environments and provides a valuable resource for AM dissection and surgical decision-making.

Keywords: Microvascular decompression (MVD) · Arachnoid Membrane · Deep Learning · Semantic Segmentation · Surgical AI

1 Introduction

Microvascular decompression (MVD) is performed to relieve neurovascular compression syndromes, most notably trigeminal neuralgia (TN) and hemifacial spasm (HFS), by repositioning or separating the offending vessels, typically the superior cerebellar artery (SCA), anterior inferior cerebellar artery (AICA), or posterior inferior cerebellar artery (PICA), from the trigeminal nerve (cranial

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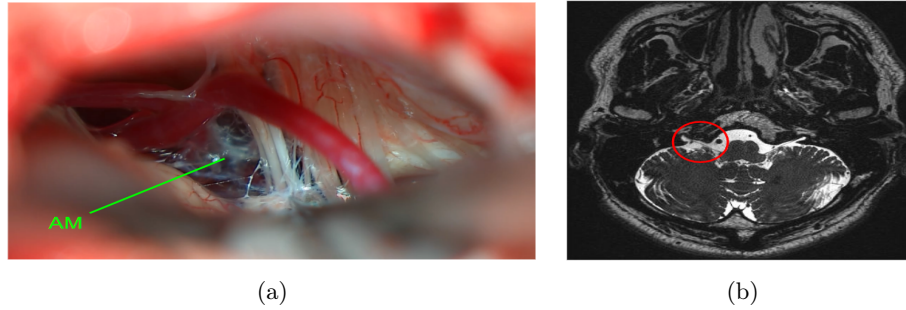


Fig. 1: Intraoperative microscopic image of the corresponding area, where the AM is clearly visible at the location indicated by the green line (a). Preoperative CISS MRI scan showing no visible AM in the region marked by a red circle (b).

nerve V) in TN or the facial nerve (cranial nerve VII) in HFS [9,16]. Sustained neurovascular contact can cause focal demyelination and abnormal signal conduction, typically at transitional zones such as the root entry zone in TN or the root exit zone in HFS, leading to neuropathic pain or involuntary movements [7]. These procedures are performed in anatomically dense operative fields, where critical neurovascular structures must be clearly visualized before safe decompression can proceed [10].

One of the main challenges in achieving clear surgical exposure is the arachnoid membrane (AM), a thin, avascular, and translucent meningeal layer that often adheres to nerves, vessels, and surrounding tissues [1,3]. To ensure both the safety and completeness of the procedure, the AM must be carefully dissected along appropriate anatomical planes [17]. This requires accurately identifying the extent and boundaries of the membrane, particularly where it partially covers or adheres to critical neurovascular structures [6]. However, AM is not discernible on the corresponding preoperative magnetic resonance (MR) images. This limitation is due to the AM's sub-voxel thickness (approximately $49\ \mu\text{m}$) and its signal similarity to cerebrospinal fluid, which together cause partial volume effects that make it effectively invisible on MRI [3,12,15,20]. This invisibility on preoperative imaging necessitates that AM dissection be performed entirely based on intraoperative visual assessment. The AM is clearly visualized under the surgical microscope in Fig.1 (a), and it is not visible in the corresponding preoperative MRI of the same region, as shown in Fig.1 (b).

As illustrated in Fig. 2, the AM (green) initially obscures the CN5 (red) and the offending artery (yellow), which become visible only after membrane dissection [10,17]. Surgeons must delineate and dissect the AM in real time under the surgical microscope or endoscope, relying solely on visual cues such as its reflectivity, subtle motion, and contrast with adjacent structures [1,15]. Inaccurate dissection planes may obscure critical anatomy or cause neurovascular injury [17], often necessitating cautious surgical progression that prolongs operative time and increases procedural risk [18]. These challenges highlight the

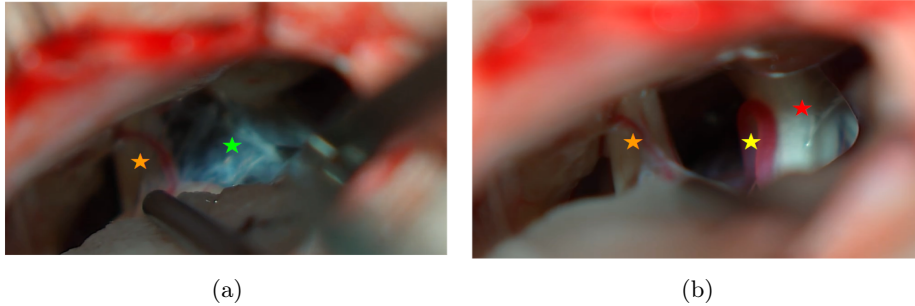


Fig. 2: Intraoperative microscopic images before (a) and after (b) AM dissection during MVD. In (a), the AM (green) overlies the surgical field, and only the CN8 (orange) is visible. After dissection in (b), the AM has been dissected, revealing the CN5 (red) and the offending artery (yellow).

need for intelligent systems that can assist with intraoperative AM localization and dissection guidance, either by enhancing the surgeon’s visual perception or through future autonomous technologies [8,11].

The application of AI to surgical vision has rapidly expanded, with growing emphasis on real-time perception, semantic segmentation of intraoperative scenes, and building the concept of the Surgeon-Machine Interface (SMI) [19]. Several studies have applied such approaches to brain surgery, including MVD, involving real-time segmentation of cranial nerves and arteries [2], boundary refinement of neurovascular structures using EnsembleEdgeFusion [5], and absolute depth estimation in intraoperative MVD scenes [13]. Despite its ubiquitous presence in brain surgeries, AM has remained relatively unaddressed in surgical AI research, probably due to its ambiguous visual boundaries and signal similarity to cerebrospinal fluid and adjacent tissues [21].

In this paper, we introduce arachnoid membrane segmentation in real-world MVD surgical scenes, laying the groundwork for future AI-assisted surgical tasks involving AM dissection. We construct a new dataset of annotated microscopic images from MVD procedures, with pixel-level AM labels manually created by expert neurosurgeons. To improve performance on this visually subtle target, we propose a query-wise AM loss that enhances segmentation by focusing supervision on AM-relevant regions. Moreover, we explore the clinical relevance of AM segmentation and its potential integration into AI-assisted surgical workflows.

2 Methodology

In this section, we present the overall methodology of our approach. We first introduce the newly constructed MVD dataset focused on AM regions. Next, we describe the baseline segmentation model architecture based on Mask2Former [4]. Lastly, we explain our proposed Query-wise AM Supervision Loss, which is designed to enhance class-specific segmentation performance.

2.1 Dataset Configuration

We constructed a microscopic MVD segmentation dataset of arachnoid membrane and adjacent anatomical structures. This dataset includes 7 patients who underwent MVD surgeries at Bongseng Memorial Hospital (BMH) performed by a single neurosurgeon between 2020 and April 2025. 5 patients were diagnosed with TN, and 2 patients with HFS. Patient ages ranged from 52 to 84 years, with 5 females and 2 males. The pathology was left-sided in 4 cases and right-sided in 3. The study was approved by the Institutional Review Board (IRB) under protocol number BSIRB-2024-008.

Surgeries were recorded using high-resolution surgical microscopes TIVATO 700 and Vario S88 (Zeiss, Jena, Germany) with videos captured at 1080p resolution and 30 FPS. For each case, representative video segments were selected based on the presence of AM dissection and neurovascular manipulation. A total of 554 frames were manually annotated by neurosurgeons, and all surgical videos were de-identified to ensure patient anonymity.

Annotations were performed based on 40 anatomical classes to capture detailed neurovascular structures, which were subsequently consolidated into 10 classes for AM segmentation by merging related structures to enhance surgical relevance and mitigate class imbalance. Cranial nerves (including CN5, CN7, CN8, CN9, and CN10) were grouped into a single CN class. Vascular structures, consisting of arteries such as AICA, PICA, and SCA, veins including the SPV, and small-caliber vessels, were consolidated into a unified vessel class. Major anatomical structures, including the cerebellum, brainstem, and tentorium, were grouped into a generalized tissue class. Tools used in surgery were categorized into surgical instruments and surgical materials. Surgical instruments included devices such as suction tips and bipolar forceps, while surgical materials included items such as teflon and cottonoids.

To enhance the model’s capacity to capture the nuanced appearance and interaction of the AM, we introduced AM-associated interaction labels. AM indicates regions where the membrane is not attached to other structures and represents the pure form of the arachnoid membrane. Depending on the attached structure, CN-AM, vessel-AM, and tissue-AM labels are assigned accordingly.

2.2 Baseline Architecture

We adopt Mask2Former [4] as our baseline segmentation architecture due to its strong performance across various semantic segmentation tasks. Mask2Former consists of a hierarchical backbone that extracts multi-scale image features, a pixel decoder that hierarchically upsamples high-resolution pixel embeddings from low-level features, and a transformer decoder that jointly predicts class labels and corresponding segmentation masks using learnable query embeddings and pixel embeddings. As the hierarchical backbone, we utilize the Swin Transformer [14], which provides robust hierarchical visual representations suitable for dense prediction. The detailed architecture of Mask2Former is depicted in Fig. 3.

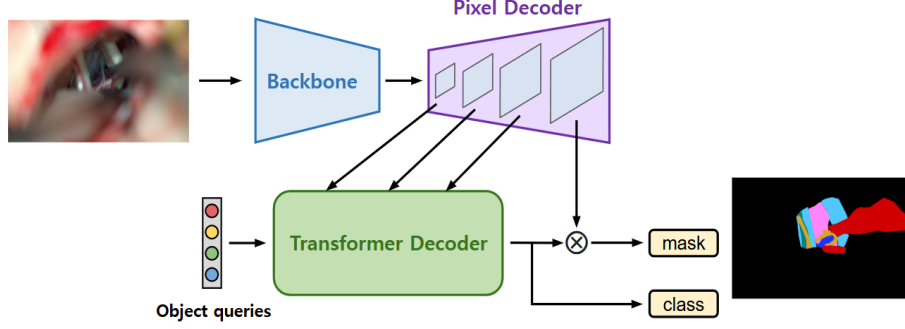


Fig. 3: Illustration of the baseline Mask2Former architecture, composed of a backbone, a pixel decoder, and a transformer decoder that predicts masks via masked attention with learnable object queries.

2.3 Query-wise AM Supervision Loss

We introduce a query-wise AM supervision loss (AM loss) to enhance segmentation sensitivity for AM regions. Due to the translucent and structurally ambiguous nature of AM, conventional loss terms often fail to provide sufficient supervision for this class. To address this, our AM loss applies direct supervision between the predicted masks and the ground-truth AM mask, encouraging at least one query embedding to specialize in AM-specific features. To focus the supervision on the most informative candidates, we compute the loss only over the top- K queries whose predicted masks exhibit the highest similarity to the AM region. The AM loss is defined as:

$$\mathcal{L}_{\text{am}} = \frac{1}{K} \sum_{q \in \mathcal{T}} \left(1 - \frac{2 \cdot \sum \hat{y}_q \cdot y_{\text{am}}}{\sum \hat{y}_q + \sum y_{\text{am}} + \epsilon} \right) \quad (1)$$

where $\mathcal{T}$ denotes the indices of the top- K queries ranked by overlap with the AM mask, $\hat{y}_q$ denotes the predicted mask from the q -th query in the final decoder layer, y_{am} is the binary ground-truth mask for the AM region, and ϵ is a small constant added for numerical stability.

The overall training objective integrates the standard Mask2Former loss, which consists of a classification loss $\mathcal{L}_{\text{cls}}$, a binary cross-entropy loss $\mathcal{L}_{\text{ce}}$, and a Dice loss $\mathcal{L}_{\text{dice}}$, with the proposed AM loss $\mathcal{L}_{\text{am}}$:

$$\mathcal{L}_{\text{m2f}} = \lambda_{\text{cls}} \cdot \mathcal{L}_{\text{cls}} + \lambda_{\text{ce}} \cdot \mathcal{L}_{\text{ce}} + \lambda_{\text{dice}} \cdot \mathcal{L}_{\text{dice}} \quad (2)$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{m2f}} + \lambda_{\text{am}} \cdot \mathcal{L}_{\text{am}} \quad (3)$$

where λ_{cls} , λ_{ce} , λ_{dice} , and λ_{am} are scalar weights for each loss component.

3 Experiments

3.1 Implementation Details

Our method was implemented using Mask2Former in the Detectron2 framework and trained on 2 NVIDIA RTX 4090 GPUs. We used a Swin-Large backbone pretrained on ImageNet-22K with a patch size of 4 and a window size of 12. The model architecture includes a multi-scale deformable attention pixel decoder and a 6-layer masked transformer decoder with 8 attention heads. Optimization was performed using AdamW with a base learning rate of $1e-4$, weight decay of 0.05, and a cosine learning rate schedule over 160k iterations. Input images had a resolution of 1920×1080 , and data augmentation included random resizing, 512×512 cropping, color jittering, and horizontal flipping.

3.2 Experimental Results

Quantitative Evaluation. We evaluated the effectiveness of the proposed AM loss on our private AM-MVD dataset using class-wise Intersection over Union (cIoU) and mean IoU (mIoU) as evaluation metrics. Table 1 presents a quantitative comparison between the baseline Mask2Former and the model trained with the proposed AM loss. The AM loss significantly improves segmentation performance on pure AM regions, increasing the IoU for the AM class from 21.09 to 28.44. The overall mIoU also increases slightly, from 33.65 to 34.40, although minor performance drops are observed in certain non-AM classes such as CN and Surgical Instrument.

Qualitative Results. Fig. 4 illustrates qualitative comparisons between the original Mask2Former and the same model trained with the proposed AM loss. The results show that incorporating the AM loss makes the model more sensitive to AM regions, leading to more precise segmentation in areas that are often missed by the baseline model. In addition, our method demonstrates strong segmentation performance in complex surgical scenes from MVD procedures, where multiple anatomical structures, such as CNs, vessels, surgical instruments, and AM regions, are densely packed and frequently overlapping. These visual results indicate that the model is capable of clearly delineating such components, highlighting its robustness in real-world intraoperative conditions.

Table 1: Quantitative comparison of semantic segmentation performance on the AM-MVD dataset showing class-wise IoU and mean IoU for the baseline Mask2Former and our method with AM Loss.

Method	Back-ground	CN	Vessel	Surgical Instrument	Surgical Materials	AM	Tissue	CN -AM	Vessel -AM	Tissue -AM	mIoU
Mask2Former (baseline)	89.11	44.91	50.53	26.12	35.51	21.09	29.40	10.37	15.80	13.68	33.65
Mask2Former (AM loss)	89.68	39.64	57.27	23.76	39.27	28.44	27.52	9.30	23.63	5.48	34.40

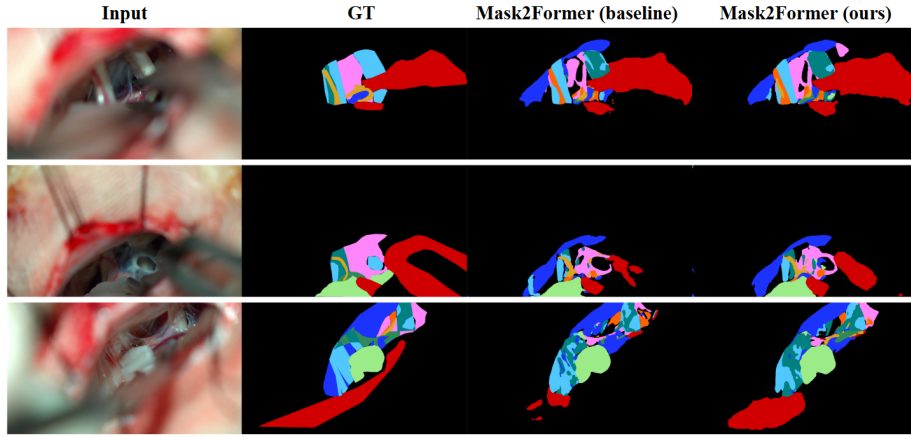


Fig. 4: Comparison of the baseline and AM loss models in MVD scenes. The AM loss model shows improved segmentation of subtle AM regions. Colors: background (black), CN (sky blue), vessels (orange), surgical instruments (dark red), surgical materials (light green), AM (pink), tissue (blue), CN-AM (blue-green), vessel-AM (gold), and tissue-AM (green).

4 Discussion

Our study introduces novel classes of AI assistance that focus on surgically important yet visually ambiguous structures. Segmentation performance on the AM notably improved when membrane-specific visual features were explicitly supervised using the proposed AM loss. The IoU for the AM class increased by 7.35%, and the model became more sensitive to its boundaries and more reliable in distinguishing it from adjacent anatomical entities. However, the model exhibited slight performance decline in other classes, including CN, surgical instruments, and composite regions such as CN-AM and Tissue-AM. These declines are likely attributable to the model’s increased emphasis on pure AM regions, which resulted in relatively reduced supervision for overlapping anatomical configurations. Despite these trade-offs, the enhanced sensitivity to membrane boundaries offers meaningful support for advancing AM dissection research and provides support for intraoperative surgical decision-making.

Limitations. The transparency of the AM can vary depending on lighting conditions, the presence of surrounding fluid, and membrane thickness. As shown in Figure 5(a), AM regions with reduced thickness and cerebrospinal fluid fluctuation were segmented less accurately than thicker, more opaque regions. In Figure 5(b), moisture-covered or brightly illuminated CN surfaces were sometimes misclassified as CN5-AM, reflecting the model’s sensitivity to surface reflectance. Figure 5(c) further shows the model’s difficulty in identifying composite configurations, such as AM encompassing CN5 (CN-AM), which were often simplified

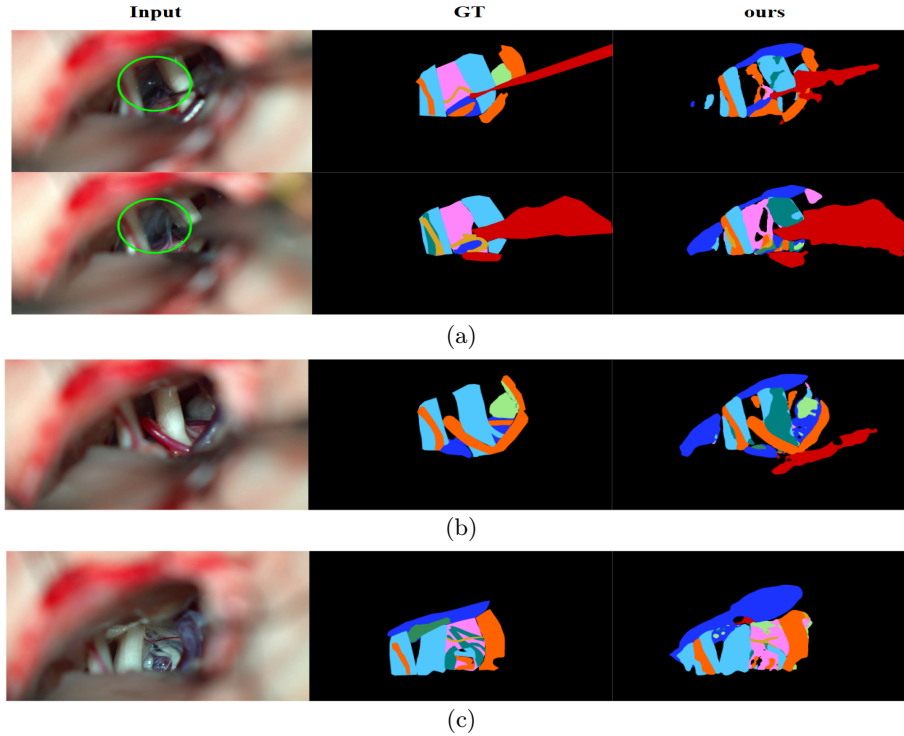


Fig. 5: Representative failure cases in AM segmentation. (a) shows reduced segmentation performance in transparent AM regions. (b) illustrates confusion with fluid-covered or highly reflective cranial nerve surfaces. (c) presents failure to distinguish composite configurations.

into standalone AM predictions. This tendency highlights persistent limitations in segmenting complex AM configurations.

5 Conclusion

In this work, we introduced the first AM segmentation framework for real-world MVD surgical scenes by constructing a pixel-level annotated dataset and proposing a query-wise AM supervision loss. Our results demonstrate improved segmentation of the arachnoid membrane, despite its subtle and variable appearance. However, challenges remain in generalizing to membrane appearances that are transparent, moist, or affected by lighting conditions. To address these limitations, we plan to improve the robustness of the model to visual variability in the appearance of AM. We are also actively expanding our dataset to include more diverse surgical cases and anatomical variations. In parallel, our goal is to extend this work toward an SMI that not only segments the AM but also supports intraoperative decision making for safe and effective arachnoid dissection.

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